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RESEARCH ARTICLE

DDNet: A Robust, and Reliable Hybrid Machine Learning Model for Effective Detection of Depression Among University Students

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ABSTRACT In recent years, the detection of depression among university students has become an increasingly critical issue. This paper presents a depression detection network (DDNet), a novel approach utilizing a three-stage stacked ensemble model to address this challenge. The proposed model incorporates two different Multilayer Perceptron (MLP) and a Stochastic Gradient Descent (SGD) as the 1st stage base classifier, an MLP, and CatBoost (CB) as 2nd stage base classifiers, with a Lasso Regressor (LASSO) serving as the meta-classifier. The hyperparameter of the used models has been optimized using random search. The optimal configuration has been determined through extensive experimentation with various machine learning (ML) models and settings to ensure high performance. Two different datasets (Dataset-1, Dataset-2) of depression detection among university students have been used to evaluate the model. The model has achieved 98.98% and 99.16% accuracy in Dataset-1 and Dataset-2, respectively. Paired t-test has been performed to ensure the statistical significance of the proposed model. To guarantee the model's transparency, SHapley Additive exPlanations (SHAP) were employed, providing interpretability of the predictive factors. Additionally, a variation of Monte Carlo Dropout (MCD) has been used to assess the uncertainty of the model's predictions, ensuring reliability. The results indicate that the proposed model is a promising tool for mental health professionals seeking effective, interpretable, and reliable solutions for early depression detection in educational settings.

INDEX TERMS Hybrid model, depression detection, explainable AI, uncertainty analysis, university students.

I. INTRODUCTION

In recent years, the identification and mitigation of mental health disorders, notably depression, have garnered heightened focus, especially within academic settings. University students are particularly vulnerable to mental health issues owing to the distinctive challenges they encounter, such as academic stress, social isolation, financial difficulties, and the transition to adulthood. If untreated, depression may lead to significant repercussions, such as academic failure, dropout, and even grave effects including self-harm or suicide. Approximately 264 million individuals globally experience depression [1]. Studies reveal that depression rates among college students are significantly higher than those in the general population, with estimates varying between 14% and 85%. Recent research highlights that students from low- and middle-income countries experience even greater prevalence rates of depression. Specifically, 43.7% of students in India, 40.9% in Pakistan, and 52.2% in Bangladesh reported feeling symptoms of hopelessness [2], [3]. It is crucial to identify

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and respond to these issues promptly in order to reduce risks and offer support to those affected. Alarmingly, more than 66% of young individuals tend to avoid discussing their mental health issues, such as depression, or seeking help for them [4], [5]. Traditional techniques of diagnosing depression, which often depend on self-report questionnaires and interviews, may be laborious, subjective, and susceptible to mistakes stemming from social stigma or hesitance to disclose symptoms. Consequently, there is an increasing need for automated, dependable, and scalable solutions for the early diagnosis of depression.

The rapid progress in artificial intelligence (AI) and ML enhances the capacity to use data-driven methodologies to address this problem [6], [7], [8]. ML models have robust predictive skills across several domains, and their use in mental health prediction has shown potential [9], [10], [11], [12]. These models, when used for behavioral and psychological data, may elucidate patterns and relationships that are difficult to discern by conventional approaches. Nonetheless, the use of ML models in mental health applications encounters several obstacles. Models must be both accurate and interpretable, since their predictions directly impact humans' well-being [13], [14], [15]. The models must manage uncertainty and provide dependable predictions to maintain credibility in sensitive applications like depression diagnosis [16], [17], [18]. This study introduces a robust and reliable hybrid ML model for the effective detection of depression in university students. The model is an innovative stacked ensemble architecture particularly designed to accurately, interpretably, and reliably identify depression in university students. DDNet employs many ML approaches organized in numerous phases to use the advantages of diverse algorithms while mitigating the shortcomings of individual models. In order to select the optimal configuration, we have experimented with various ML models, namely Logistic Regression (LR), Naive Bayes (NB), Decision Tree (DT), Random Forest (RF), Extra Tree (ET), Gradient Boosting (GB), Support Vector Machine (SVM), Extreme Gradient Boosting (XGB), Light Gradient Boosting Machine (LGBM), CB, LASSO, SGD, and MLP. The model integrates two different MLP (MLP-1, MLP-2) and SGD as the initial base classifiers, an additional MLP(MLP-3) and CB as the secondary classifiers, and LASSO as the final classifier. This ensemble framework enables the model to attain elevated prediction accuracy by consolidating the outputs of many learners, thereby enhancing generalization and mitigating overfitting. The model's configuration was refined with Random Search. This hyperparameter tuning technique facilitates rapid exploration of an extensive hyperparameter space without the computing costs associated with exhaustive search techniques such as Grid Search. Through meticulous fine-tuning of the model via comprehensive testing with various ML models and configurations, we achieved an ideal equilibrium between performance and efficiency.

Although accuracy is important, it is equally essential for the model to provide interpretable and dependable predictions, especially in the delicate field of mental health. To guarantee interpretability, we used SHAP, a well recognized model-agnostic explanation method that elucidates the importance of each feature to the model's predictions. SHAP elucidates the principal aspects affecting the model's decision-making, so assuring that stakeholders, including mental health experts, can comprehend and trust the forecasts. Furthermore, MCD was used to evaluate the model's uncertainty. Uncertainty quantification is essential in health-related applications since it offers insights into the reliability of forecasts, enabling practitioners to proceed with caution when required. A crucial research element is the statistical confirmation of DDNet's efficacy. A paired t-test was performed to evaluate the proposed model's performance against other prevalent ML models. This test verifies that the performance improvements noted in DDNet are statistically significant and not attributable to random variation. We have utilized two publicly available datasets for depression detection among university students. These datasets consist of survey-based responses, including demographic information and mental health assessment scores from validated psychological scales. Dataset-1, derived from [12], contains 804 responses with 30 features, incorporating socio-demographic and career-related stress. Dataset-2, obtained from [19], consists of 672 responses with 53 features, including socio-demographic, multiple psychological assessment scales, and mental health assessment questions. These datasets provide a diverse and representative sample of university students' mental health conditions, enabling robust model evaluation.

The significant contributions of our research work are:

1) Development of DDNet, a resilient and dependable hybrid ML model for the early identification of depression in university students, using an innovative stacked ensemble architecture that integrates MLP, SGD, CB, and LASSO, refined via Random Search for optimal performance.
2) The model attained good accuracy and performance on two separate depression detection datasets, with 98.98% accuracy on Dataset-1 and 99.06% accuracy on Dataset-2. The enhancements in model performance were shown to be statistically significant by a paired t-test, verifying that the observed improvements are not attributable to random chance. This underscores the efficacy of DDNet in reliably producing high-quality predictions for depression diagnosis, making it a potential instrument for practical mental health applications.
3) Enhancing model interpretability and transparency via SHAP to elucidate feature significance, offering insights into the determinants affecting the model's predictions, essential for utilization in sensitive fields such as mental health.

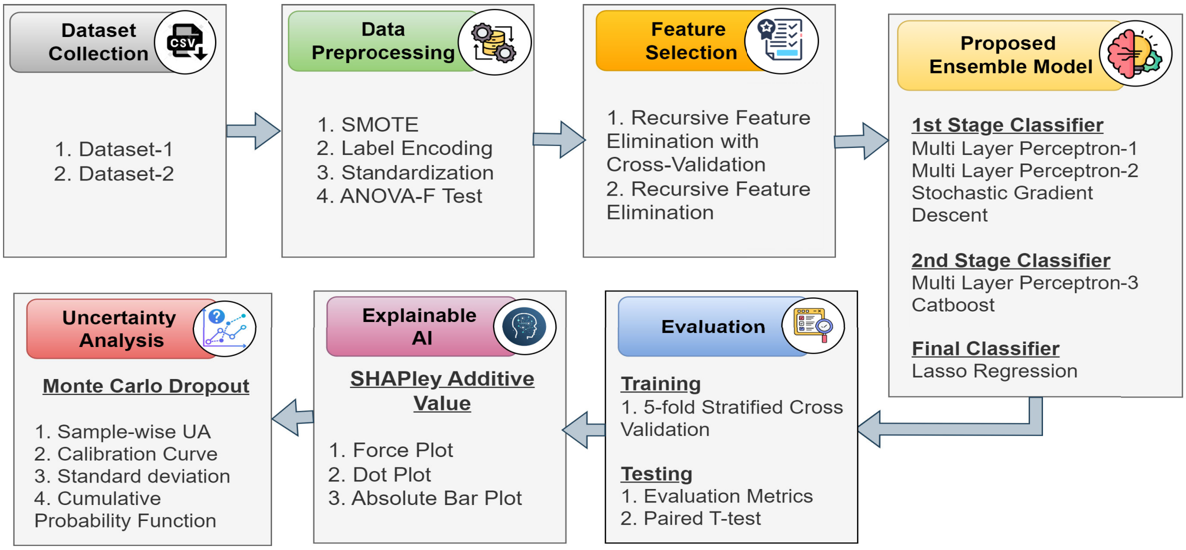

FIGURE 1. This figure presents the step-by-step workflow of the proposed work. The workflow has begun with the collection of publicly available datasets. Then the datasets are preprocessed, and then feature selection has been performed. After that, the proposed ML model has been built, trained, and evaluated. Lastly, the model has been interpreted using XAI, and uncertainty has been handled through MCD.

4) Integration of uncertainty estimates using MCD to assess the confidence in the model's predictions, guaranteeing that the model delivers outputs that are not only precise but also dependable and credible, which is essential for practical mental health applications.

II. LITERATURE REVIEW

The mental health of university students has become a significant concern, with studies indicating that young adults in academia experience worse mental health outcomes than their peers in other professions [20], [21], [22]. Meda et al. [23] conducted a study involving 1,388 students, found that nearly 20% reported severe depressive symptoms or suicidal ideation, with economic worry significantly associated with depression. The RF algorithm demonstrated high accuracy in predicting students' well-being, though it struggled to predict symptom deterioration, emphasizing the importance of cognitive and somatic symptoms in risk assessment. To address the lack of practical predictive tools for mental distress, Zhang et al. [24] developed an advanced AI application to forecast severe mental health challenges among college students. An extensive study involving 2,088 participants utilized various ML models, with the extreme GB machine achieving the highest AUC of 0.932 and demonstrating superior performance across multiple evaluation metrics. The successful deployment of this web-based AI tool highlights its potential for timely intervention and preventative strategies. University students are frequently subjected to high stress levels, leading to mental health disorders such as anxiety, depression, and burnout. Sutter et al. [25] created an ML model to forecast psychological distress using solely ecological data, employing eight classification methods on a sample dataset. The preliminary results are encouraging, indicating that with further refinement, an accurate and dependable model can be achieved. Purwandari et al. [26] concentrated on predicting Internet addiction disorder among youth through ML, aiming to enable early intervention. They matched input data to participants' responses on the GHQ-12 questionnaire, categorizing them based on their overall health status. Baird et al. [27] utilized ML techniques to explore the connection between psychological stress in refugee children and specific features of their artwork captured digitally. Their results indicated that the drawing features identified by ML corresponded with recognized clinical links between specific artistic characteristics and psychological distress.

Gonzalo et al. [28] aimed to detect these symptoms through a specially designed questionnaire and employed an ensemble of classifiers for prediction, including RF and artificial neural networks. Results indicated that the algorithms effectively identified the presence of these disturbances with high accuracy and low false negatives, demonstrating the utility of ML in monitoring student mental health. The prevalence of depression among university students has gained significant attention, especially post-COVID-19, which heightened mental health challenges linked to increased mobile technology use and remote learning. Siraji et al. [29] conducted a study at an International University in Bangladesh and employed ML algorithms to automate the detection of depression, achieving accuracy improvements of 3-5% with feature selection techniques like Chi-square and Recursive Feature Elimination (RFE), and a notable enhancement when using SparsePCA with the CB classifier. Ahmed and Ahmed [30] aimed to develop a minimalistic depression identification system that rapidly retrieves app usage data from the past week, demonstrating effective detection of depression using a diverse set of ML models, including linear, tree-based, and neural networks. This approach facilitated the identification of depressed and non-depressed students, with feature importance assessed using a stable approach and SHAP explanations enhancing model interpretability. Elovanio et al. [31] explored the application of ML techniques in extensive health research, utilizing the RF (RF) model to assess the predictive validity of various psychological scales, including the GHQ-12, Beck Depression Inventory, and the Mental Health Index. Their findings indicate that while ML can significantly improve data analysis, achieving high predictive accuracy for specific outcomes, such as the utilization of mental health services, remains a significant hurdle. A comprehensive review by Lee and Ham [32] emphasizes the adaptability of different ML approaches, such as LOR, RF, SVM, and artificial neural networks (ANN), in diagnosing depression across various data formats. This study highlights the necessity of choosing suitable ML models that correspond to the data's characteristics to maximize performance metrics like accuracy, sensitivity, and specificity.

Saha et al. [33] proposed an ensemble hybrid model combining SVM and MLP that underscores the need for robust automated detection systems. This model achieves an accuracy of 99.39% while effectively addressing class imbalance through the SMOTE technique. This model demonstrates the potential of integrating psychological and sociodemographic factors in depression detection. Recognizing the critical need for efficient screening methods, Mumenin et al. [12] utilized the General Health Questionnaire-12 alongside various ML techniques, revealing that approximately 60% of participants experienced depressive symptoms. The Extremely Randomized Tree model emerged as the most effective, achieving 90.26% accuracy and highlighting the interplay of socio-demographic variables and stressors affecting student mental health. Additionally, Bieliński et al. [34] investigated how ML algorithms contribute to the development of a virtual mental health index, assessing various ML models on a clinical dataset in terms of training duration, operational runtime, and regression accuracy. Lastly, Mohammad and Siddiqui [35] employed RF classifiers and regressors, along with hyperparameter optimization strategies, to collect mental health data through three online questionnaires, creating a distinct model for each dataset. Alam et al. [36] analyzed the perceived factors contributing to depression and suicidal ideation among Bangladeshi university students and identified academic pressure, loneliness, and financial issues as key stressors. Utilizing the Apriori association algorithm, this study provided insights into the complex relationships between these factors and the risk of suicide, emphasizing the importance of preventive measures. Munir et al. [37] focused on identifying predictors of depression among university students through a classification model that utilized a dataset collected via social media. The stacking classifier with 24 attributes showed the highest accuracy, underscoring the need for effective models to detect depression early and promote mental health.

III. METHODOLOGY

This task comprises a number of important steps. The comprehensive workflow is illustrated in figure 1.

A. DATASET

Two different datasets were used to evaluate this study. We used the data from one of our previous papers [12] for the first dataset. The second dataset was collected from a journal paper and made publicly available [19]. The first dataset (Dataset-1) has 804 responses having 30 feature columns, and 1 target column, while the second dataset (Dataset-2) has 672 responses having 53 feature columns and 1 target column. Dataset-1 contains 412 negative and 260 positive cases, while Dataset-2 contains 486 negative and 318 positive cases. Dataset-1 contains the Socio-demographic questions (SD1-SD9), Career-and Job-seeking stress-related questions (C1-C11), and General Health Questionnaire (G1-G12). Dataset-2 contains the Socio-demographic questions (SD1-SD9), Patient Health Questionnaire (PHQ), Pittsburgh Sleep Quality Index (PSQI), GAD (Generalized Anxiety Disorder), and UCLA Loneliness Scale-8 questions.

B. DATA PREPROCESSING

Data preparation is essential in the development of ML models. In real-world scenarios, data often suffers from issues such as incompleteness, inconsistency, inaccuracy, and missing critical attribute values. The process of data preparation encompasses the cleaning, organizing, and formatting of raw data to ensure it is appropriate for ML applications. In this study, we have utilized several data preparation techniques, including:

1) Categorical Data Encoding: Category data encoding refers to the transformation of categorical variables into numerical formats, enabling their use in ML models. The values in the datasets were categorical response from the participants. Thus, label encoder has been implemented to convert them into numerical values.
2) Class Imbalance Handling: Synthetic Minority Oversampling Technique (SMOTE) is an advanced resampling method that generates synthetic instances of the minority class rather than merely duplicating existing samples. This method prevents overfitting, a risk when using basic oversampling techniques, and improves the model's ability to generalize across different class distributions. We analyzed the class distribution in both datasets and observed class imbalance. SMOTE was then applied to create synthetic samples for the minority class, ensuring that both classes were more evenly represented. After applying SMOTE, there was an equal number of positive and negative cases

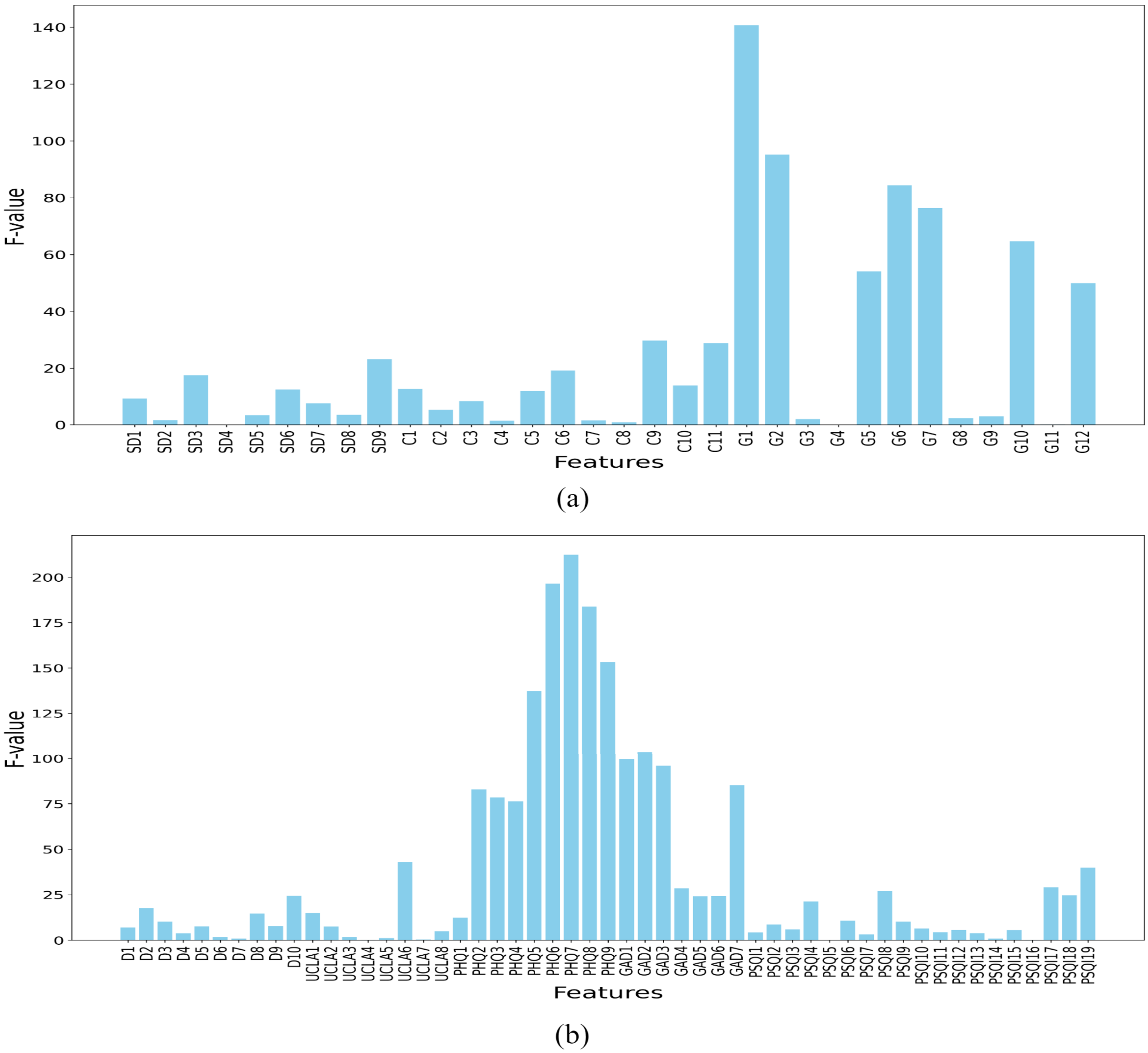

FIGURE 2. **This figure shows the graph of feature importance generated using ANOVA-F test for (a) Dataset-1, and (b) Dataset-2 respectively. The correlation among the features is understood through the figure.**

in Dataset-1 (486 positive and 486 negative) and Dataset-2 (412 positive and 412 negative).

3) Feature Significance Calculation: Understanding the significance of each piece of information in relation to the target variable is essential when constructing predictive models. ANOVA-F test has been applied for finding the significant features from the datasets. Figure 2(a) depicts the F-values of characteristics associated with socio-demographic parameters (SD1 to SD9) and questions from the GHQ-12 (General Health Questionnaire) of Dataset-1. Among the socio-demographic characteristics, SD9, SD7, and SD1 have comparatively greater significance. The GHQ-12 components are predominant in this study, with G1, G2, G6, and G7 exhibiting the greatest F-values. This underscores that answers to the GHQ-12 inquiries are essential for forecasting depression among university students. Figure 2(b) displays the F-values for several variables, including UCLA, PHQ, GAD, and PSQI scales, as well as demographic data (D1 to D10) of Dataset-2. The Patient Health Questionnaire (PHQ) has the greatest F-values, with PHQ6, PHQ7, PHQ8, and PHQ9 demonstrating the most significant impact, underscoring their essential role in predicting depression. Furthermore, elements from the GAD (Generalized Anxiety Disorder) scale, including GAD3 and GAD4, have significant relevance, emphasizing the correlation between anxiety and depression. The

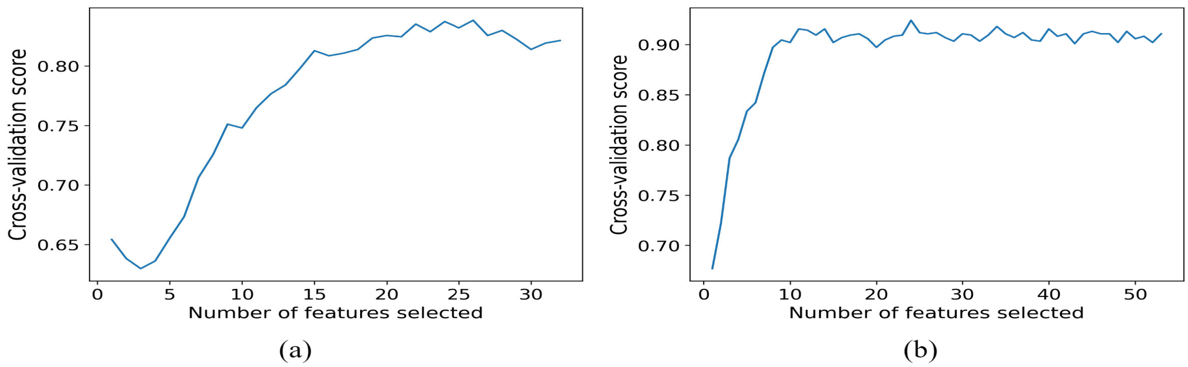

FIGURE 3. This figure presents the selection of optimal features using RFECV for (a) Dataset-1, and (Dataset-2) respectively.

UCLA Loneliness Scale has moderate significance, with UCLA4 being particularly notable. Factors affecting sleep disruption evaluated by the Pittsburgh Sleep Quality Index (PSQI) contribute to the analysis but often show lower F-values relative to other scales.

4) Dimensionality Reduction: Recursive Feature Elimination with Cross-Validation (RFECV) was used to identify the best subset of features, hence improving the performance and robustness of the proposed model. RFECV was used with RF, employing a 5-fold StratifiedKFold to maintain balanced class distributions across the folds. The main goal was to decrease dimensionality while preserving the most relevant characteristics, thereby enhancing generalization and alleviating overfitting. The RFECV procedure in the first dataset determined an ideal count of roughly 26 features, as seen in the figure 3(a). The cross-validation score initially rose markedly with the addition of features, but after 26 features, the performance stabilized. This indicates that more variables beyond this juncture may introduce noise instead of enhancing predictive capability. In the second dataset, the RFECV determined an ideal number of roughly 35 features, as seen in the figure 3(b). The model's performance improved significantly with the addition of features, with decreasing returns beyond 35 features. This constant result demonstrates the efficacy of the chosen feature subset in identifying the fundamental patterns of the dataset. The feature selection technique decreased model complexity, allowing concentration on the most useful characteristics, thereby enhancing accuracy and dependability while reducing computing expenses.

C. PROPOSED ENSEMBLE MODEL

Stacking, which is also referred to as stacked generalization using the ensemble method, operates on a straightforward principle: rather than relying on basic functions like the voting ensemble, the predictions of all predictors in a group are combined. One advantage of stacking is its ability to leverage the performance of several high-performing models in a classification or regression task, resulting in predictions that surpass the performance of any individual model within the ensemble [38], [39], [40]. In this study, we propose a three-stage ensemble model (EM) or hybrid framework that synergistically combines multiple base classifiers with a meta-classifier. The overarching goal is to leverage the diverse strengths of various classifiers to enhance the model's predictive performance. Figure 4 presents the architecture of the proposed model. Algorithm 1 presents the detailed pseudocode for the proposed method.

1) FIRST STAGE OF THE ENSEMBLE MODEL

In the first stage of the proposed ensemble model, three models (MLP-1, MLP-2, and SGD) are used as the base classifiers. The hyperparameters of the models are optimized using random search and are presented in table 1. The models are trained in parallel using five-fold cross-validation, and their predictions are combined through stacking to generate meta-features for the subsequent stage. The methodology for the first stage is described step by step as follows:

Let $\mathbf{X}_{\text{train}} \in \mathbb{R}^{n\times m}$ represent the training dataset, where n is the number of samples, and m is the number of features. The corresponding target labels are represented as $\mathbf{y}_{\text{train}} \in \mathbb{R}^{n}$. Similarly, $\mathbf{X}_{\text{test}}$ denotes the test dataset with unknown labels. Before training the models, we applied a standard scaling transformation to each feature to ensure that all features contribute equally to the model training. The standardized features, $\mathbf{X}_{\text{scaled}}$, are computed as:

$$\mathbf{X}_{\text{scaled}} = \frac{\mathbf{X} - \mu}{\sigma} \tag{1}$$

where μ is the mean and σ is the standard deviation of the feature values.

The training dataset was split into five folds using StratifiedKFold cross-validation, ensuring that the target variable distribution $\mathbf{y}_{\text{train}}$ was preserved across the folds. For each fold $k \in \{1, \ldots, 5\}$, the training set was split into a training subset $(\mathbf{X}^{k}_{\text{train\_fold}}, \mathbf{y}^{k}_{\text{train\_fold}})$ and a validation subset $(\mathbf{X}^{k}_{\text{val\_fold}}, \mathbf{y}^{k}_{\text{val\_fold}})$. Each model was cloned and trained on the training subset, and predictions were made on

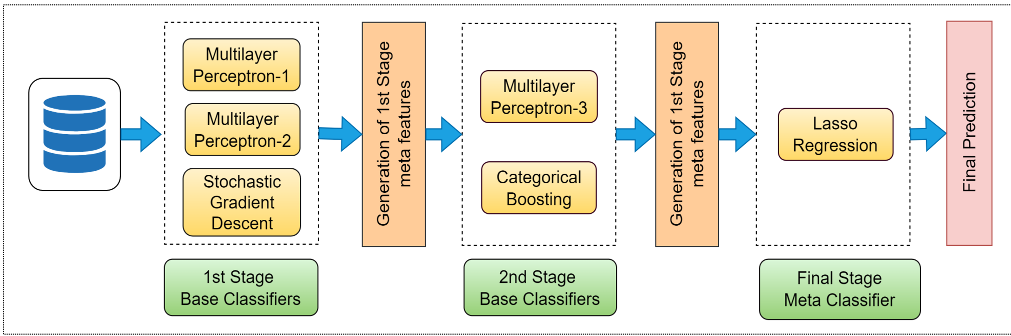

FIGURE 4. **The architecture of the proposed hybrid model. It consists of three different stages, namely the first, second and final stage. At first, the data is passed through the first stage classifiers. The predictions from the base classifiers in combined to generate the meta features. Then the meta features are passed through the second stage base classifiers. From the prediction of the classifiers, the second stage meta features are generated. Finally, the features are passed through the final classifier.**

the validation subset:

$$\hat{\mathbf{y}}^k_{\text{val}} = f(\mathbf{X}^k_{\text{val\_fold}}) \tag{2}$$

where f represents the model's function after training.Finally, The predicted values $\hat{\mathbf{y}}^k_{\text{val}}$ were stored in the meta-feature matrix $\mathbf{X}^k_{\text{meta\_train}}$ for each validation fold. After all folds, the average cross-validation accuracy for each model was calculated as:

$$\text{Accuracy} = \frac{1}{K}\sum_{k=1}^{K} \text{Accuracy}(\mathbf{y}^k_{\text{val\_fold}}, \hat{\mathbf{y}}^k_{\text{val}}) \tag{3}$$

where $K = 5$ is the number of folds.

After completing the cross-validation, predictions from all models were averaged across all folds to generate meta-features for the test dataset:

$$\mathbf{X}^i_{\text{meta\_test}} = \frac{1}{K}\sum_{k=1}^{K} \hat{\mathbf{y}}^k_{\text{test}} \tag{4}$$

where $\hat{\mathbf{y}}^k_{\text{test}}$ represents the predictions made by model i on the test dataset during fold k. This stacking procedure resulted in a meta-feature matrix $\mathbf{X}_{\text{meta\_train}}$ for the training set and $\mathbf{X}_{\text{meta\_test}}$ for the test set, with each column corresponding to the output of one of the models in stage 1.

D. SECOND STAGE OF THE ENSEMBLE MODEL

In the second stage of the model, the meta-features generated from the first stage are used as input to train another set of ML models. These models are stacked to further refine the predictions, and the resulting meta-features are passed on to the final stage. Two different ML models (MLP-3, and CB) were employed in this stage to process the meta-features generated in Stage 1. The hyperparameters of the models are optimized using random search and are presented in table 1. Each of these models was trained using the meta-features produced in the first stage of the model. The aim is to leverage the meta-level representation of the data to capture complex patterns and interactions that the base-level models may not fully represent.

Just as in the first stage, five-fold StratifiedKFold cross-validation was applied to ensure a balanced representation of the target variable $\mathbf{y}_{\text{train}}$ across the folds. For each fold $k \in \{1, \ldots, 5\}$, the meta-feature training set from the first stage ($\mathbf{X}_{\text{meta\_train}}$) was split into training and validation subsets. Specifically, ($\mathbf{X}^k_{\text{train\_fold}}, \mathbf{y}^k_{\text{train\_fold}}$) and ($\mathbf{X}^k_{\text{val\_fold}}, \mathbf{y}^k_{\text{val\_fold}}$). Then, each model was cloned and trained on the training subset of meta-features, and predictions were made on the validation subset:

$$\hat{\mathbf{y}}^k_{\text{val}} = f(\mathbf{X}^k_{\text{val\_fold}}) \tag{5}$$

where f represents the function of the model after training. Lastly, the predicted values $\hat{\mathbf{y}}^k_{\text{val}}$ were stored in the meta-feature matrix $\mathbf{X}^k_{\text{meta\_train\_stage2}}$ for each validation fold. After completing all folds, the average accuracy for each model was calculated as:

$$\text{Accuracy} = \frac{1}{K}\sum_{k=1}^{K} \text{Accuracy}(\mathbf{y}^k_{\text{val\_fold}}, \hat{\mathbf{y}}^k_{\text{val}}) \tag{6}$$

where $K = 5$ represents the number of folds.

Once the cross-validation process was completed, predictions for the test dataset were generated by averaging the predictions across all folds:

$$\mathbf{X}^i_{\text{meta\_test\_stage2}} = \frac{1}{K}\sum_{k=1}^{K} \hat{\mathbf{y}}^k_{\text{test}} \tag{7}$$

where $\hat{\mathbf{y}}^k_{\text{test}}$ are the predictions made by model i on the test set during fold k. This stacking process generated the second-stage meta-feature matrices $\mathbf{X}_{\text{meta\_train\_stage2}}$ for the training data and $\mathbf{X}_{\text{meta\_test\_stage2}}$ for the test data, which serve as input to the final stage of the ensemble model.

E. THIRD STAGE OF THE ENSEMBLE MODEL

In the third and final stage of the proposed model, the meta-features generated from the second stage are used as input to a regression model, which produces the final predictions. The objective of this stage is to aggregate the

TABLE 1. Hyperparameter optimization using random search.

Model	Search Parameters	Optimized Parameters
MLP-1	Layer_sizes: [(32, 16), (64, 32), (64, 32)] learning_rate: ['constant', 'adaptive'] alpha: [0.0001, 0.001]	hidden_layer_sizes: (128, 64, 32) max_iter: 1000 learning_rate: 'constant' alpha: 0.0001 random_state: 42
MLP-2	Layer_sizes: [(64, 32), (128, 64), (256, 128)] learning_rate: ['constant', 'adaptive'] alpha: [0.0001, 0.001]	hidden_layer_sizes: (128, 64, 32) max_iter: 1000 learning_rate: 'constant' alpha: 0.0001 random_state: 42
SGD	max_iter: [1000, 2000] tol: [1e-4, 1e-3] loss: ['hinge', 'log'] penalty: ['l2', 'l1', 'elasticnet']	max_iter: 1000 tol: 1e-3 loss: 'log' penalty: 'l2' random_state: 42
CB	iterations: [50, 100, 200] depth: [4, 6, 8] learning_rate: [0.01, 0.1, 0.2]	iterations: 100 depth: 6 learning_rate: 0.1 verbose: 0 random_state: 42
MLP-3	Layer_sizes: [(64, 32), (128, 64), (256, 128)] learning_rate: ['constant', 'adaptive'] alpha: [0.0001, 0.001]	hidden_layer_sizes: (64, 32) max_iter: 1000 learning_rate: 'constant' alpha: 0.0001 random_state: 42
LASSO	alphas: [0.1, 0.01, 0.001] cv: [3, 5, 10] max_iter: [1000, 2000]	cv: 5 alphas: [0.001] max_iter: 1000 random_state: 42

information provided by the meta-features and generate the final classification outcome.

The final model used in this stage is a LASSO model with cross-validation (LassoCV). LR applies L_1 regularization to the coefficients, which helps in selecting a subset of important features from the second-stage meta-features. The regularization term is defined as:

$$\text{Lasso Objective} = \frac{1}{2n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda\sum_{j=1}^{m}|w_j| \tag{8}$$

where n is the number of samples, $\hat{y}_i$ are the predicted values, w_j are the model coefficients, and λ is the regularization parameter controlling the strength of the penalty.

In this case, LASSO was used, which automatically selects the optimal value of λ by performing five-fold cross-validation on the meta-feature training set. This ensures that the model balances the trade-off between minimizing the prediction error and enforcing sparsity in the feature set. The meta-features generated from the second stage, $\mathbf{X}_{\text{meta\_train\_stage2}}$, are used to train the LASSO model. Specifically, the model fits the training data using:

$$\hat{\mathbf{y}}_{\text{train}} = f_{\text{Lasso}}(\mathbf{X}_{\text{meta\_train\_stage2}}) \tag{9}$$

where f_{Lasso} represents the LASSO function and $\hat{\mathbf{y}}_{\text{train}}$ are the predicted values on the training set. After training the model, predictions are made on the meta-feature test set $\mathbf{X}_{\text{meta\_test\_stage2}}$:

$$\hat{\mathbf{y}}_{\text{test}} = f_{\text{Lasso}}(\mathbf{X}_{\text{meta\_test\_stage2}}) \tag{10}$$

Since LASSO is a continuous model, the predicted values $\hat{\mathbf{y}}_{\text{test}}$ are continuous in nature. Therefore, a rounding function is applied to convert the continuous predictions into binary class labels:

$$\hat{\mathbf{y}}_{\text{binary}} = \text{round}(\hat{\mathbf{y}}_{\text{test}}) \tag{11}$$

The resulting $\hat{\mathbf{y}}_{\text{binary}}$ values represent the final predicted class labels for the test set.

TABLE 2. Performance of the proposed model on the datasets.

Metric	Dataset 1	Dataset 2
Accuracy	0.9898	0.9916
Precision	0.9918	0.9917
Recall	0.9875	0.9916
F1 Score	0.9896	0.9916
Log Loss	0.5368	0.0450
AUC	0.9898	99.24
Cohen's Kappa	0.9664	0.9832

The use of LASSO in the final stage ensures that the model is not only robust but also interpretable, as the L_1 regularization drives the coefficients of less important features to zero, effectively selecting the most relevant meta-features. Thus, the proposed model is precise, accurate, and transparent.

We have experimented with various ensemble architectures, including 2-stage and 4-stage models. However, the 3-stage architecture emerged as the most effective configuration, providing superior performance without excessive computational overhead.

TABLE 3. Experimental combinations of models across different stages. Accuracy is written as "Acc".

Configuration	Stage 1	Stage 2	Stage 3	Stage 4	Acc (Dataset-1) (%)	Acc (Dataset-2) (%)
2-Stage	RF, ET, SVM	CB, XGB	-	-	96.82	97.14
2-Stage	MLP-1, MLP-2	RF, GB	-	-	97.12	97.45
2-Stage	NB, DT, LGBM	CB, LASSO	-	-	96.35	96.80
2-Stage	RF, SGD, SVM	XGB, ET	-	-	96.90	97.21
3-Stage (Proposed)	MLP-1, MLP-2, SGD	CB, MLP	LASSO	-	98.98	99.06
3-Stage	DT, NB, RF	GB, SVM	LGBM	-	97.21	97.68
3-Stage	LR, SVM, SGD	ET, RF	XGB	-	97.75	98.12
3-Stage	MLP-1, NB, ET	LGBM, RF	CB	-	97.88	98.23
3-Stage	LGBM, RF, SVM	CB, ET	LASSO	-	98.45	98.79
3-Stage	LR, SGD, MLP	GB, RF	XGB	-	98.02	98.40
3-Stage	DT, RF, MLP-2	LGBM, CB	ET	-	98.61	98.98
3-Stage	LR, NB, ET	XGB, GB	RF	-	97.95	98.31
3-Stage	SGD, SVM, MLP-1	CB, RF	LASSO	-	98.25	98.65
4-Stage	LR, NB	DT, ET	CB, XGB	LASSO	98.12	98.15
4-Stage	RF, ET	SVM, GB	CB, LGBM	LASSO	98.08	98.11
4-Stage	RF, DT	MLP-1, SGD	CB, ET	XGB	98.05	98.07
4-Stage	LR, NB	RF, SVM	CB, GB	LASSO	98.90	98.95
4-Stage	MLP-1, NB, ET	LGBM, SGD	RF, XGB	CB	98.02	99.04
4-Stage	DT, SGD, LR	RF, ET	CB, GB	XGB	98.01	99.03
4-Stage	RF, MLP-2	SVM, LGBM	CB, GB	XGB	98.04	98.06
4-Stage	LR, ET	DT, SGD	CB, XGB	LASSO	98.96	98.99

1) HYPERPARAMETER OPTIMIZATION

We have used random search to tune the hyperparameters of the proposed model. Random Search provides a more efficient alternative to grid search by randomly selecting from the specified hyperparameter space, enabling it to investigate a wider array of parameter combinations without the need of painstakingly testing every potential configuration. This methodology is particularly advantageous for intricate models such as ours, where optimizing many models inside a stacked architecture might be resource-intensive. Random Search has been used to optimize essential hyperparameters for each model, including the number of hidden layers, learning rate, alpha (regularization term), maximum iterations, depth, and number of estimators. In the MLP, hyperparameters such as hidden_layer_sizes and learning_rate were optimized to enhance the capture of non-linear relationships in the data, whereas in the SGD, parameters like iterations and loss were adjusted to improve model stability and mitigate variance. Likewise, CB Classifier was refined for its iterations, learning rate, and depth to enhance its efficacy on tabular data, while LassoCV was adjusted to guarantee the regularization of the final model. Table 1 presents the models, the random search parameters, and optimized parameters found using random search.

The random search method assessed several hyperparameter combinations using cross-validation to guarantee generalization and mitigate overfitting. This facilitated the identification of optimal configurations for each model, hence enhancing accuracy and stability. This tuning method was important in the model's exceptional overall performance.

IV. EVALUATION

A. EXPERIMENTAL SETUP

The experiments were conducted on a machine equipped with an Intel Core i5 CPU (6 cores, 4.10 GHz), 16 GB of RAM, and an NVIDIA GTX 3050 GPU. The training and evaluation processes for Dataset-1 and Dataset-2 took approximately 35 seconds and 38 seconds, respectively. These timings indicate that the proposed model is computationally efficient and can be executed on modest hardware configurations.

B. RESULT ANALYSIS

The proposed three-stage ensemble model was evaluated on two datasets using multiple performance metrics, including accuracy, precision, recall, F1 score, log loss, AUC, and Cohen's Kappa. The results for both datasets are presented in Table 2. The model attained an accuracy of 98.98%, a precision of 99.18%, a recall of 98.75%, and an F1 score of 98.96% for Dataset-1. The results indicate that the model exhibited consistent performance in precision, recall, and total accuracy, indicating its effective balance between false positives and false negatives. The F1 score, as a harmonic mean of accuracy and recall, indicates that the performance is dependable in both identifying genuine positives and minimizing false positives. The log loss for Dataset 1 was 0.5368, suggesting that the model's probability estimations were well calibrated. The AUC of 0.9898 indicates that the model demonstrated good discriminative ability, efficiently distinguishing between the positive and negative classes. Cohen's Kappa, an agreement metric that adjusts for chance, was 0.9664, indicating a very high degree of concordance between the predicted and actual labels, hence reinforcing trustworthiness. The model exhibited superior performance

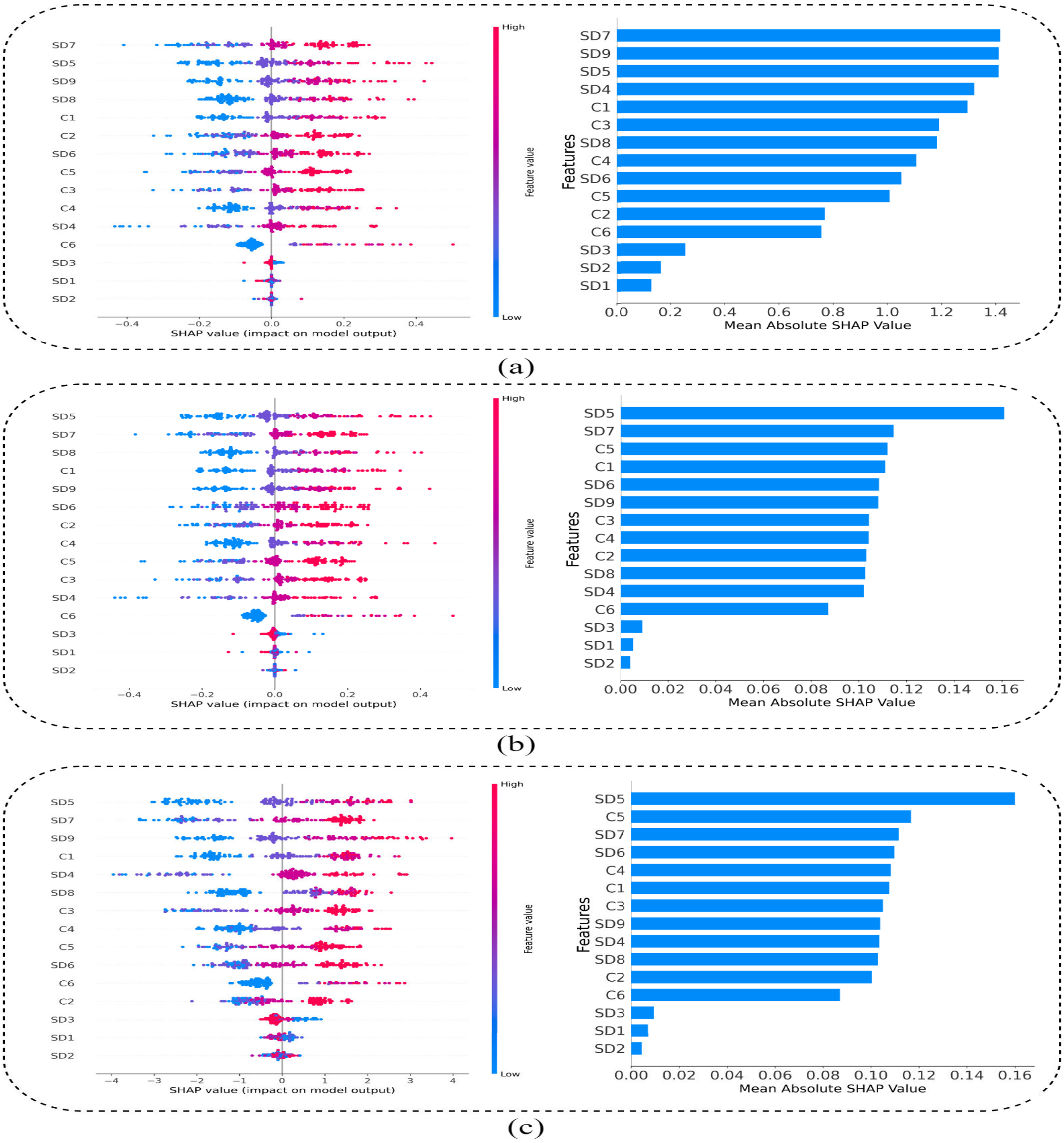

FIGURE 5. The figure presents dot plots (left) and bar plot (right) for the base classifiers (a) MLP-1, (b) MLP-2, and (c) SGD on Dataset-1.

on Dataset 2, achieving an accuracy of 99.16%, precision of 99.17%, recall of 99.16%, and an F1 score of 99.16%. These measures demonstrate the ability in predicting both positive and negative classes. The log loss was very minimal at 0.0450, indicating extremely confident and precise probability estimates. The AUC was reported as 99.24, highlighting the exceptional capacity to differentiate between the two groups with considerable confidence. Cohen's Kappa was very high at 0.9832, supporting substantial concordance between predictions and actual labels, even when adjusting for chance.

The high accuracy, precision, recall, and F1-scores obtained by the model across both datasets indicate that it is highly effective at detecting depression among university students. Additionally, the low log loss values demonstrate that the probability estimates are reliable, while the high AUC and Cohen's Kappa scores further confirm its robustness and reliability. These results suggest that the proposed model can be a reliable tool for predicting depression with high confidence and consistency, making it well-suited for practical applications in mental health analysis.

C. RATIONALE FOR 3-STAGE MODEL

Our proposed framework uses a 3-stage architecture, which is chosen based on a combination of empirical experimentation

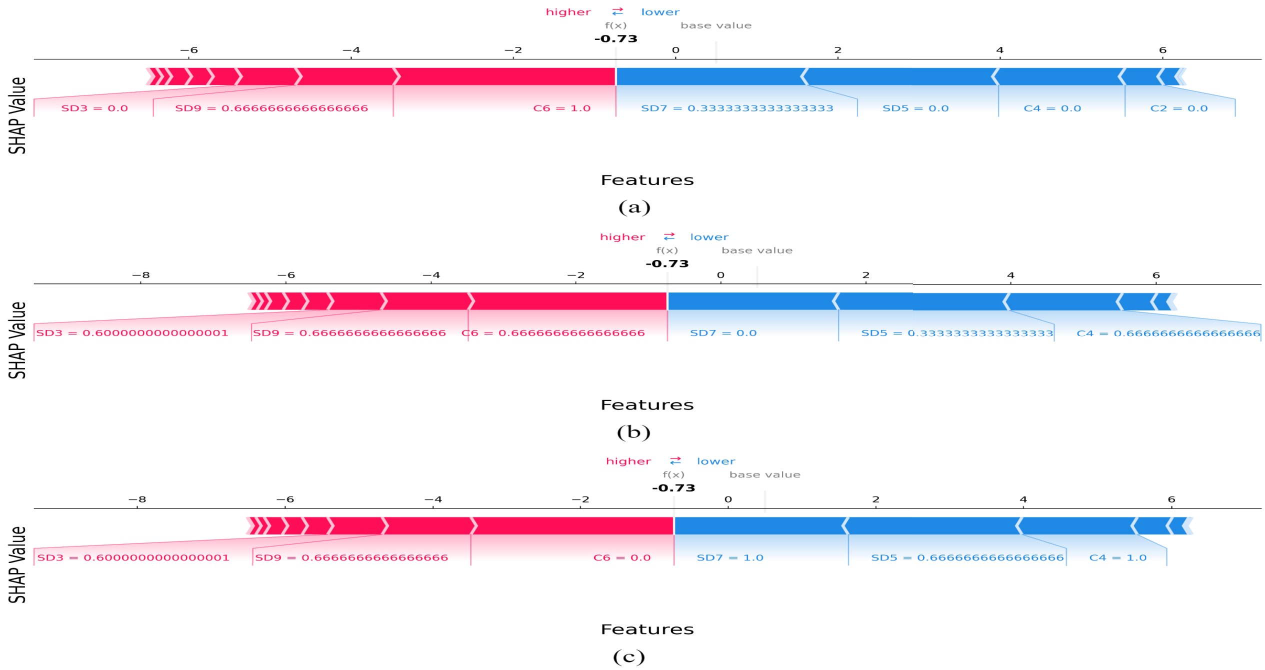

FIGURE 6. The figure presents force plot generated for a particular sample for the base classifiers (a) MLP-1, (b) MLP-2, and (c) SGD on Dataset-1.

and theoretical considerations aimed at achieving an optimal balance between model complexity, performance, and interpretability. We experimented with various ensemble architectures, including 2-stage and 4-stage models. However, the 3-stage architecture emerged as the most effective configuration during our experiments, providing superior performance without excessive computational overhead. Table 3 provides several combinations and their obtained results in both datasets.

In a 2-stage model, we found that combining the base classifiers' outputs directly into a single meta-classifier lacked sufficient refinement. This resulted in lower accuracy than the 3-stage model, as it limited its ability to capture complex patterns and interactions within the data. On the other hand, increasing the architecture to four stages introduced additional layers of complexity, which led to diminishing returns. The performance improvement was marginal and did not justify the increased computational cost and risk of overfitting. To arrive at the 3-stage design, we conducted experiments comparing configurations (2)-stage, 3-stage, and 4-stage) on both datasets. The results consistently showed that the 3-stage model achieved the best balance between accuracy, interpretability, and computational efficiency. For instance, while the 4-stage model marginally improved accuracy in some cases, it increased training time significantly without adding meaningful insights.

D. STATISTICAL SIGNIFICANCE ANALYSIS

In this study, we conducted a statistical significance test to compare the performance of the proposed model against several existing ML models. The aim is to ascertain the statistical significance of the performance enhancement offered by the proposed model in comparison to other conventional models. The paired t-test served as the principal significance test, comparing the performance characteristics of the models on an identical dataset. This test determines whether the observed variations in model performance are attributable to chance or signify a true improvement. The paired t-test was performed by comparing essential performance indicators (including accuracy, F1-score, and AUC-ROC) for each model across numerous iterations. A p-value threshold of 0.05 was established for statistical significance. Should the p-value go below 0.05, we reject the null hypothesis in favor of the alternative, signifying that the enhancement in our model's performance is statistically significant. Table 4 provides the statistical comparison among the proposed and existing ML models. The results of the statistical significance test indicate that the proposed model surpasses the current ML methods. The p-values for all tests were below 0.05, indicating that the enhancements in accuracy, precision, recall, F1-score, and AUC-ROC attained by our model are statistically significant. This indicates that the observed disparities are improbable to result from random chance and that our ensemble model offers a significant enhancement over conventional classifiers. The paired t-test was a critical step for validation, ensuring confidence in the proposed model's results.

E. COMPARISON WITH EXISTING ML MODELS

The proposed model exhibits enhanced efficacy in identifying depression in university students relative to many commonly

TABLE 4. Comparison of the proposed model with Other ML Models on Dataset-1 and Dataset-2.

Model	Dataset-1					Dataset-2				
	Acc	F1-Score	AUC	p-value(Acc)	p-value(AUC)	Acc	F1-Score	AUC	p-value (Acc)	p-value (AUC)
Proposed	**0.98**	**0.98**	**0.99**	-	-	**0.99**	**0.98**	**0.99**	-	-
LR	0.89	0.88	0.89	0.0002	0.0004	0.90	0.89	0.90	0.0003	0.0005
NB	0.87	0.85	0.88	0.0001	0.0003	0.88	0.86	0.88	0.0002	0.0004
DT	0.88	0.87	0.88	0.0003	0.0005	0.89	0.88	0.89	0.0004	0.0006
RF	0.93	0.92	0.93	0.001	0.002	0.95	0.94	0.95	0.0015	0.0022
ET	0.93	0.92	0.94	0.0015	0.0021	0.94	0.93	0.94	0.0017	0.0023
GB	0.94	0.93	0.95	0.0012	0.0018	0.95	0.94	0.96	0.0018	0.0021
SVM	0.91	0.90	0.92	0.0005	0.0008	0.92	0.91	0.92	0.0007	0.001
XGB	0.94	0.92	0.95	0.003	0.004	0.95	0.95	0.96	0.004	0.005
LGBM	0.93	0.95	0.92	0.002	0.003	0.97	0.95	0.95	0.0025	0.0035
CB	0.95	0.96	0.97	0.0025	0.0032	0.95	0.94	0.95	0.003	0.0038
LASSO	0.90	0.89	0.90	0.0004	0.0006	0.91	0.90	0.91	0.0005	0.0007
SGD	0.95	0.95	0.95	0.0007	0.0010	0.95	0.95	0.95	0.0009	0.0012
MLP	0.93	0.92	0.94	0.0013	0.0017	0.94	0.93	0.94	0.0016	0.0019

used ML models. Table 4 demonstrates the results obtained by the proposed and other ML models in Dataset-1 and Dataset-2.

Among the individual models, CB had superior performance, achieving 95% accuracy, 0.95 F1-score, and 0.94 AUC-ROC on Dataset-1, and 95% accuracy, 0.95 F1-score, and 0.98 AUC-ROC on Dataset-2. Nevertheless, these scores remain inferior to the proposed model, suggesting that DDNet's ensemble architecture is more adept at identifying the patterns linked to depression symptoms in these datasets. Other high-performing models, such as LGBM and XGB, had commendable results, attaining up to 93% accuracy and 0.94 AUC-ROC on Dataset-2. Models characterized by simpler architectures, including LR and NB, exhibited reduced accuracy scores, fluctuating between 87-89% across both datasets, accompanied by proportionally diminished F1-scores and AUC-ROC values. While these models are quick and uncomplicated, their restricted complexity hinders their capacity to discern intricate correlations within the data, particularly for difficult tasks such as depression identification. Similarly, tree-based models like DT and RF attained accuracy ratings ranging from 88% to 94%, exhibiting enhanced performance compared to simpler classifiers, although still falling short of the predictive capability shown by the proposed model. Multi-stage ensemble models, such as DDNet, which integrate the capabilities of diverse classifiers at different stages, demonstrate a significant benefit. The integration of MLP, SGD, and CB models facilitates the amalgamation of varied insights from distinct classifiers, hence enhancing the generalizability and robustness of its predictions. This stratified methodology enables the proposed model to use both linear and non-linear decision boundaries, hence improving accuracy and dependability. In conclusion, DDNet surpassed individual ML models across both datasets, underscoring the efficacy of its ensemble design.

F. COMPARISON WITH EXISTING WORKS

Table 5 provides a comparative analysis of various methodologies for mental health detection, focusing on their accuracy and F1-score metrics. The proposed model achieves an outstanding accuracy of 98.98% and 99.16% in two different datasets, positioning it as a leading approach in this domain. In comparison, Zhang et al. [24] reported an accuracy of 85.0% with their XGB, while Gonzalo et al. [28] achieved 96.9% using an ensemble of RF and ANN. Siraji et al. [29] attained an impressive accuracy of 99.78%, reinforcing the reliability of ML methods. Saha et al. [33] utilized a hybrid SVM and MLP ensemble, reaching 99.39% accuracy and an F1-score of 99.51%. In contrast, Sabbir Ahmed and Nova Ahmed [30] obtained an accuracy of 82.4% with LGBM, and Mumenin et al. [12] achieved 90.26% with ET, although their F1-scores were not reported. Lastly, Munir et al. [37] employed a stacking classifier, resulting in a lower accuracy of 77.45%. Overall, this comparison underscores the advancements in ML applications for mental health detection and highlights the superior performance of the proposed model.

TABLE 5. Comparison of existing works on depression detection.

Reference	Method	Accuracy	F1-Score
Proposed	Three Stage Ensemble	99.86%	99.86%
Zhang et al. [24]	XGB	85.0%	-
Gonzalo et al. [28]	RF and ANN	96.9%	-
Siraji et al. [29]	ML Algorithms	99.78%	99.78%
Sabbir Ahmed [30]	LGBM	82.4%	78.4%
Saha et al. [33]	Hybrid SVM and MLP	99.39%	99.51%
Mumenin et al. [12]	ET	90.26%	-
Munir et al. [37]	Stacking Classifier	77.45%	-

V. EXPLAINABLE AI

We have used explainable AI (SHAP) to improve the interpretability of the proposed model. SHAP offers a cohesive framework for elucidating the outputs of ML models by attributing a contribution value to each feature. This facilitates a clear comprehension of the influence of different

Algorithm 1 Pseudocode for the Proposed Model

Require: Training data X_{train}, labels y_{train}, test data X_{test}, cross-validation folds k, number of models n in ensemble
Ensure: Predicted labels for X_{test}

Stage 1: Base Classifier Training and Prediction
for each model $m_1, m_2, \ldots, m_n$ in first-stage base classifiers (MLP-1, MLP-2, SGD) **do**
  for each fold $(X_{train}^{(fold)}, y_{train}^{(fold)})$ in k-fold cross-validation on X_{train} **do**
    Train base classifier m_i on $X_{train}^{(fold)}, y_{train}^{(fold)}$
    Predict on validation fold to obtain out-of-fold predictions $P_{train}^{(m_i)}$
    Predict on X_{test} to obtain test predictions $P_{test}^{(m_i)}$
  end for
end for
Aggregate predictions from all models in Stage 1 to form meta-features X_{meta1} for training and X_{meta1_test} for testing
Stage 2: Second-Level Classifier Training and Prediction
for each model m'_1, m'_2 in second-stage classifiers (MLP, CB) **do**
  for each fold $(X_{meta1}^{(fold)}, y_{train}^{(fold)})$ in k-fold cross-validation on X_{meta1} **do**
    Train second-stage classifier m'_i on $X_{meta1}^{(fold)}, y_{train}^{(fold)}$
    Predict on validation fold to obtain out-of-fold predictions $P_{meta1}^{(m'_i)}$
    Predict on X_{meta1_test} to obtain test predictions $P_{meta1\_test}^{(m'_i)}$
  end for
end for
Aggregate predictions from all models in Stage 2 to form meta-features X_{meta2} for training and X_{meta2_test} for testing
Stage 3: Meta-Classifier Training and Prediction
Train LASSO as meta-classifier on X_{meta2}, y_{train}
Predict on X_{meta2_test} using meta-classifier to obtain final predictions y_{pred}
Output Predicted labels y_{pred} with uncertainty estimates $= 0$

variables on predictions, enhancing the interpretability and reliability of the model's decision-making process. The SHAP analysis was performed on the three base models in our ensemble—MLP-1, MLP-2, and SGD—illustrating the influence of each feature on the model's output. SHAP dot plots have shown the analysis, and bar charts for each model, demonstrating the impact of different variables on model predictions. Also, SHAP force plot has been provided to explain rationale behind the prediction of a particular sample.

A. XAI OF DATASET-1

Figure 5 presents SHAP dot plots and bar charts for each model, illustrating the distribution of SHAP values in the dot plots and the overall feature relevance in the bar charts. The SHAP analysis for the first model (MLP-1), presented in figure 5(a) indicates that traits such as SD7, SD9, and SD5 are the most significant. The dot plot on the left illustrates the distribution of SHAP values for all examples, with each point denoting a feature's contribution to the prediction of a specific instance. The color gradient transitioning from blue to red signifies the feature's value, ranging from low to high. For example, SD7 and SD9 have a significant positive impact when their values are elevated, steering the predictions towards the positive class. The bar chart on the right illustrates the mean absolute SHAP values, indicating the relative significance of each characteristic. C1, C3, and SD4 have significant effect, but their contributions are rather little. In the second model (MLP-2) shown in figure 5(b), the SHAP values indicate a somewhat different pattern of feature significance. Although SD5 and SD7 continue to be primary contributors, the feature C5 has emerged as one of the most significant. The dot plot illustrates that elevated feature values for SD5 (red) drive predictions towards the positive class, whereas diminished values tend to attract predictions towards the negative class. The bar chart prioritizes the significance of characteristics such as C5, C1, and SD6, each of which is essential in influencing the model's predictions. The distribution of SHAP values across instances in this model is more concentrated than in MLP-1, indicating more consistency in feature relevance across samples. The SHAP analysis for the SGD model, shown in figure 5(c) identifies SD5, C5, and SD7 as the most influential characteristics affecting the model's predictions. The SHAP values for SD5 exhibit a much wider range than the other models, with substantial positive contributions for elevated values and considerable negative contributions for diminished values. The bar chart on the right highlights the preeminence of SD5 in the SGD model, with a mean SHAP value markedly superior than all other features. Other significant elements include C5, SD6, and C4, however their total influence is somewhat less relative to SD5. Although feature SD1 is rated lower, it nevertheless influences the model's decision-making, as seen by its modest but significant SHAP values.

Figure 6(a), 6(b), and 6(c) presents the force plot for a particular sample generated from MLP-1, MLP-2 and SGD respectively. In all three instances, SD3, SD9, and C6 exhibit substantial positive SHAP values, elevating the prediction to a greater magnitude. In figure 6(a), SD9 (value of 0.666) and C6 (value of 1.0) are the most influential, enhancing the forecast, but SD7 and SD5 provide negative contributions, diminishing the prediction. In figure 6(b), SD9 and C6 significantly contribute to increasing the forecast, but SD7 and SD5 provide a negative counterbalance. In figure 6(c), SD3 and SD9 elevate the forecast, but C4 and SD5 provide negative SHAP values, diminishing the prediction. The model illustrates how a mix of factors that contribute favorably and adversely influences the final forecast across all plots. The SHAP values clearly reveal the characteristics influencing the model's decisions, with SD3, SD9, and C6 constantly

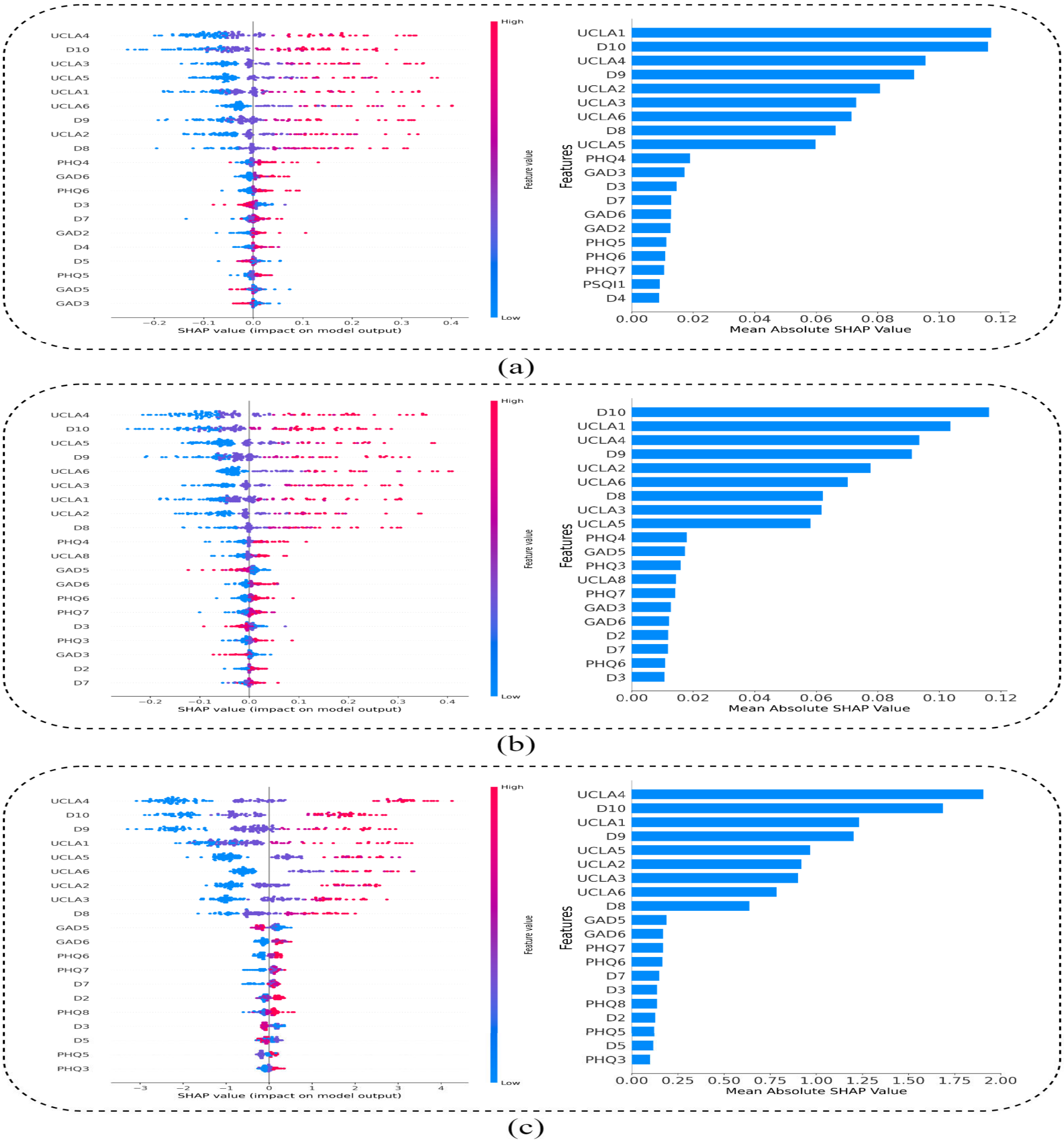

FIGURE 7. **The figure presents dot plots (left) and bar plot (right) for the base classifiers (a) MLP-1, (b) MLP-2, and (c) SGD on Dataset-2.**

exerting significant impact on elevating the forecasts in these cases. The force charts provide a clear and comprehensible method to elucidate the model's decision-making process for each sample.

In all three models, traits such as SD5, SD7, and C5 consistently emerge as primary contributors to the forecasts. This signifies their strength in affecting the model results, irrespective of the model type. The fundamental traits remain consistent, however their significance differs across models. For example, SD5 assumes a much more pivotal function in the SGD model compared to MLP-1 or MLP-2. C5 has increased significance in MLP-2 and SGD relative to MLP-1. Elevated values of SD7, SD5, and C1 consistently drive predictions towards the positive class, while diminished levels have an opposing influence on the predictions. This interaction illustrates how specific feature values directly influence the model's output, enhancing interpretability.

B. XAI OF DATASET-2

Figure 7 presents the violin chart and bar chart explaining the base classifiers on Dataset-2. In the MLP-1 model, presented in figure 7(a), SHAP values revealed that characteristics like UCLA4, D10, and UCLA1 had the most substantial influence on predictions. UCLA4 repeatedly shown a significant beneficial impact when its value was elevated, steering the model towards a favorable prediction. The dot plot illustrated

FIGURE 8. The figure presents force plot generated for a particular sample for the base classifiers (a) MLP-1, (b) MLP-2, and (c) SGD on Dataset-2.

the contribution of individual cases with varying feature values to the model output, but the bar chart of mean absolute SHAP values offered a comprehensive ranking of feature significance. Features of lesser significance, such as GAD3 and D4, had little influence on the predictions. In MLP-2, presented in figure 7(b), the SHAP values exhibited a same trend, but with minor variations in feature ranking. D10, UCLA1, and UCLA4 continued to be the most significant characteristics, but the impact of other features, including PHQ4 and GAD5, became more pronounced relative to MLP-1. The distribution of SHAP values across cases demonstrated that elevated levels of D10 and UCLA1 favorably influenced the model's output, whereas diminished values inclined predictions toward the negative class. In the SGD model, presented in figure 7(c), SHAP values had a more significant influence on the primary characteristics. UCLA4, D10, and D9 had the greatest impact on the model's predictions. The SHAP values for UCLA4 in SGD had a more pronounced and consistent impact across instances than those of the MLP models. Furthermore, attributes such as GAD5 and PHQ3 gained increased significance in this model, indicating that SGD depended more on these characteristics than the neural network models.

Figure 8(a), 8(b), and 8(c) presents the force plot for a particular sample generated from MLP-1, MLP-2 and SGD respectively. The SHAP force graphs demonstrate the contribution of specific characteristics to the model's predictions. In all three cases ((a), (b), and (c)), the characteristic D8 has the most substantial influence, constantly driving the prediction lower with considerable negative SHAP values (-6.93 in (a) and (b), and -7.75 in (c)). This signifies that D8 has the most substantial effect on these predictions, diminishing the model's output. Conversely, characteristics such as UCLA4, UCLA1, UCLA5, and UCLA6 have minimal contributions, shown by their near-zero SHAP values, indicating that they do not significantly influence the model's decisions for these samples. The uniformity throughout the plots underscores the significance of D8 in influencing the result, while the other characteristics have no influence. This study illustrates the model's transparency and interpretability, providing insight into the relative significance of certain factors in particular forecasts.

In all models, traits such as UCLA4, D10, and UCLA1 were consistently identified as the primary contributors to forecasts, underscoring their significance in forecasting the target variable. Despite the presence of identical elements in several models, the degree of their impact differed. For example, UCLA4 had a greater SHAP value in the SGD model compared to the MLP models, indicating that each model assigns varying importance to features. Elevated levels of UCLA4 and D10 often inclined the predictions towards the positive class, while diminished values frequently led to a negative prediction.

While SHAP is a powerful tool for interpreting ML models, its application to highly non-linear and complex architectures, such as the proposed stacked ensemble model, has certain limitations that warrant discussion. In ensemble methods, where predictions result from the combined outputs of multiple base models, SHAP values are computed independently for each model and aggregated to represent

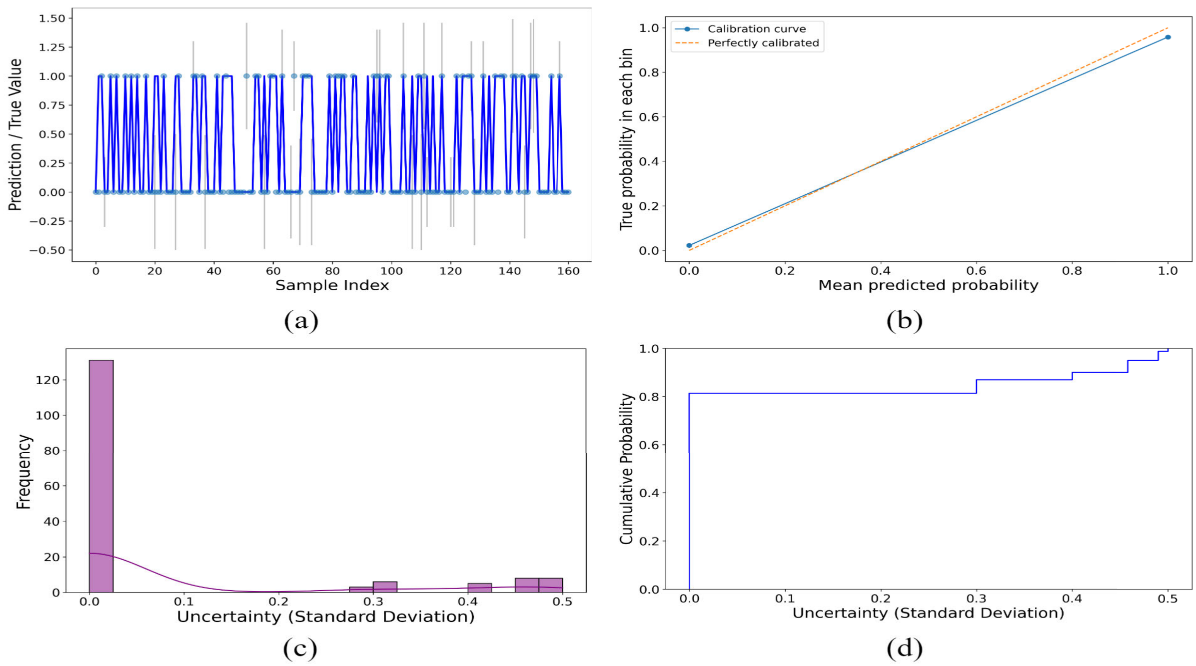

FIGURE 9. Uncertainty analysis on Dataset-1 using (a) Sample-wise uncertainty plot, (b) Calibration curve, (c) Standard deviation of the probability of uncertainty, and (d) Cumulative probability function.

overall feature importance. However, this aggregation can obscure the nuanced differences in feature significance across individual models, potentially leading to oversimplified interpretations. For instance, significant features in one base model might have minimal impact on another, making the aggregated SHAP values less reflective of individual model behaviors. While SHAP provides a quantitative measure of feature importance, its interpretations must be contextualized. For example, highly influential features identified through SHAP analysis may still require domain expertise to understand their practical implications in mental health applications. To address these limitations and enhance the interpretability of the proposed model, SHAP analysis was conducted separately for each base model to isolate the feature contributions specific to each model. This provides a clearer understanding of the role individual features play at different stages of the ensemble.

VI. UNCERTAINTY ANALYSIS

We have used a strategy akin to MCD, training several models with varied initializations to capture prediction variability. This method enables us to measure the uncertainty linked to the model's predictions, offering insights into the model's confidence on an individual sample basis. We have established an ensemble of 10 LassoCV models, each configured with a distinct random seed. The use of many instances of the final model brings little unpredictability into the predictions. Each model has been trained on the meta-learned features produced by prior stages of the ensemble, and predictions have been conducted on the test set. The forecasts from all ten models have been aggregated for each test sample, and the mean prediction was calculated to get the final prediction. The standard deviation of the predictions among the models has been computed as the uncertainty metric for each sample. The uncertainty quantification procedure yields a five-fold output, with the mean forecast derived by averaging the outputs of all models, which becomes the final prediction. The uncertainty (Standard Deviation) is determined by the prediction variability across the models, indicating the model's confidence level. A low standard deviation signifies strong confidence, while a more considerable number implies more uncertainty in the forecast. Visualizations, including uncertainty plots, histograms, and scatter plots, were used to illustrate the distribution of uncertainty and its correlation with the predictions made by the model. This method facilitates a more transparent comprehension of confidence of the model in its predictions and potential unreliability, improving the system's resilience.

A. UNCERTAINTY ANALYSIS ON DATASET-1

Figure 9 provides a detailed overview of the uncertainty analysis results of Dataset-1. Figure 9(a) illustrates the predicted values in conjunction with the actual values for each sample, while the vertical lines denote the uncertainty (standard deviation) of the predictions. The model's predictions vary between 0 and 1, reflecting the binary nature of the categorization challenge. The uncertainty bars signify that, for most samples, the uncertainty is limited, since the standard deviation of forecasts is negligible, indicating a high confidence level in the predictions. In some circumstances, the elevated

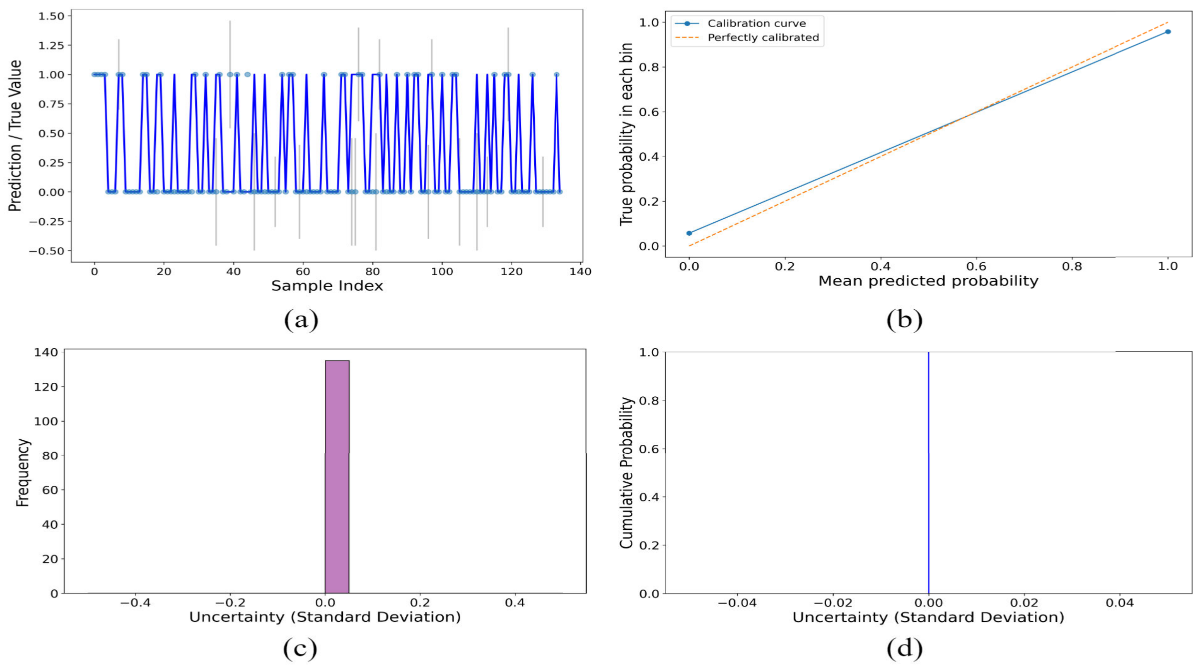

FIGURE 10. Uncertainty analysis on Dataset-2 using (a) Sample-wise uncertainty plot, (b) Calibration curve, (c) Standard deviation of probability of uncertainty and (d) Cumulative probability function.

uncertainty values suggest that the model exhibits less confidence in its predictions, highlighting borderline occurrences. Figure 9(b) presents the calibration curve, juxtaposing the anticipated probabilities with the actual probabilities inside each bin. An accurately calibrated model would closely align with the dotted line representing perfect calibration, as seen in this instance. The model's calibration curve closely adheres to the diagonal line, signifying that the projected probabilities correspond effectively with the actual observed probabilities, so affirming that the model is well-calibrated and yields dependable probability estimates. Figure 9(c) illustrates the distribution of uncertainty (standard deviation) across all forecasts. The histogram indicates that most of the predictions exhibit little uncertainty, characterized by a pronounced peak at zero. This indicates that most forecasts are made with considerable confidence. A limited number of occurrences demonstrate increased uncertainty, indicating that there are some circumstances in which the model exhibits less confidence in its predictions. Figure 9(d) illustrates the cumulative probability of uncertainty, indicating the fraction of occurrences that are below each uncertainty threshold. The sharp ascent in the curve signifies that almost 80% of forecasts exhibit little uncertainty (close to zero), so corroborating the assertion that the model has substantial confidence in most of its predictions. A limited number of examples have elevated uncertainty levels, indicating difficult or unclear circumstances for the model.

B. UNCERTAINTY ANALYSIS ON DATASET-2

Figure 10 provides a detailed overview of the uncertainty analysis results of Dataset-2. Figure 10(a) illustrates the anticipated values juxtaposed with the actual values, while the vertical bars denote uncertainty. In the majority of samples, the uncertainty approaches zero, signifying the model's substantial confidence in these predictions. Several examples exhibit more uncertainty, indicating complex scenarios where the confidence in the result is diminished. figure 10(b) depicts the calibration curve, which juxtaposes the projected probabilities against the actual probabilities. The diagonal dashed line signifies a model that is completely calibrated. The calibration curve closely aligns with the ideal line, indicating that the anticipated probabilities correspond effectively with the actual observed probabilities. This indicates that the model is accurately calibrated and yields dependable probability estimates. Figure 10(c) illustrates the distribution of uncertainty for the predictions in histogram form. The zero bar signifies that the majority of predictions demonstrate no uncertainty, indicating that the model has high confidence for almost all samples. This conclusion suggests that the model exhibits high confidence in its predictions, with no ambiguous circumstances characterized by significant uncertainty. Figure 10(d) illustrates the cumulative likelihood of uncertainty. The figure ascends steeply at zero, indicating that the model exhibits excellent confidence in almost all of its predictions, which possess negligible uncertainty. The pronounced increase in cumulative probability suggests the absence of substantial uncertainty, reinforcing the predictive dependability.

VII. CONCLUSION

This research introduces DDNet, a resilient and dependable hybrid ML model aimed at the early identification of depression in university students. The proposed model employs a three-stage stacked ensemble methodology, including two MLPs and SGD as initial classifiers, an additional MLP and CB as secondary classifiers, and LASSO as the meta-classifier. The model was optimized by random search, resulting in exceptional performance, attaining 98.98% accuracy on Dataset-1 and 99.16% accuracy on Dataset-2. In addition to its excellent accuracy, a variant of MCD was used to evaluate prediction uncertainty, to assure the dependability of the predictions. The use of SHAP facilitated interpretability by elucidating the contribution of each feature to the predictions made by the model. The proposed model is both precise and transparent, which is essential for its use in depression detection. The statistical significance of the performance of the model, validated by a paired t-test, demonstrated that DDNet's enhancements over conventional ML models were not attributable to chance but constituted a substantial development.

Notwithstanding its advantages, the proposed model has certain restrictions. The model was first trained and validated using two publicly accessible datasets, which may not comprehensively reflect the variety of actual student populations. The datasets may be deficient in detailed socio-demographic factors, thereby affecting the model's generalizability across many situations, including pupils from various cultural or geographical origins. Another issue is that the proposed model concentrates on binary categorization (presence or absence of depression), which may not adequately encompass the range of depression severity. Mental health disorders, such as depression, occur on a continuum, and future models may include multi-class categorization to discern different degrees of depression severity. Numerous scopes for further study exist to enhance the influence of this work. Initially, it would be advantageous to use DDNet over a wider array of datasets, including those with varied socio-demographic characteristics and cultural contexts, to evaluate the model's generalizability. This may include gathering fresh data from other student demographics or extending the model to encompass other educational settings, such as secondary schools or vocational institutions. Furthermore, further research might investigate the integration of other mental health disorders, such as worry or stress, into the model, therefore establishing a multi-modal system proficient at concurrently recognizing diverse psychological states. This may result in the creation of an extensive mental health monitoring instrument for educational environments.

In conclusion, DDNet is a promising and efficient instrument for identifying depression in university students. With more refining, increased generalizability, and greater interpretability, it has the capacity to substantially assist in mental health monitoring and early intervention initiatives within educational settings.

CONFLICT OF INTEREST

There is no conflict to declare.

DATA AVAILABILITY STATEMENT

Data will be made available on request.

AUTHOR CONTRIBUTION

Nasirul Mumenin: Conceptualization, Formal Analysis, Investigation, Methodology, Data Collection, Data Analysis, Writing: Original Manuscript, Writing: Revised Manuscript, Method Development; Mohammad Abu Yousuf, Madini O. Alassafi, Muhammad Mostafa Monowar, Md. Abdul Hamid: Supervision, Validation;

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